Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	14 July 2026
Team ID	
Project Name	Book a Doctor
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

R No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	* Registration through Form * Registration through Gmail * Registration through LinkedIn
FR-2	User Confirmation	* Confirmation via Email * Confirmation via OTP
FR-3	Profile Management	* Profile Setup (Clinic Address, Specialty, Qualifications, and Fees) * Document/License Upload for Admin Verification
FR-4	Search & Filter Engine	* Doctor Search by Name or Specialty * Multi-Parameter Filtering (Consultation Fees, Location, Rating)

R No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		* Real-time Status Indicator (Available / Off-duty)
FR-5	Scheduler & Calendar Configuration	* Calendar Setup (Weekly Consulting Slots configuration) * Block-out Calendar Dates (Vacations & custom off-duty times)
FR-6	Booking & Rescheduling	* Slot Selection and Booking Transaction Process * ACID-compliant Concurrent Reservation Locking (Prevents double-booking) * Rescheduling and Manual Cancellation Workflow

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.Non-Functional RequirementDescriptionNFR-1

Usability

* Responsive client dashboards built with Tailwind CSS.

* Intuitive search and workflow navigation allowing patients to complete bookings in under 3 clicks.

NFR-2

Security

* Role-based access controls using JSON Web Tokens (JWT).

* Secure password hashing using bcrypt and HTTPS encryption for data in transit.

NFR-3

Reliability

* 100% prevention of simultaneous slot reservations (double-bookings).

* Atomic transaction wrapping via ACID-compliant database sessions.

NFR-4

Performance

* Search and filter page load times kept under 1.5 seconds.

* Booking confirmations processed within 300ms using Redis query caching.

NFR-5

Availability

* Minimum platform uptime target of 99.9% outside of scheduled maintenance.

* Multi-AZ data replication for rapid failover recovery.

NFR-6

Scalability

* Microservices-based decoupled design built on the MERN stack.

* Containerized app architecture managed by Kubernetes to scale services based on peak demands.